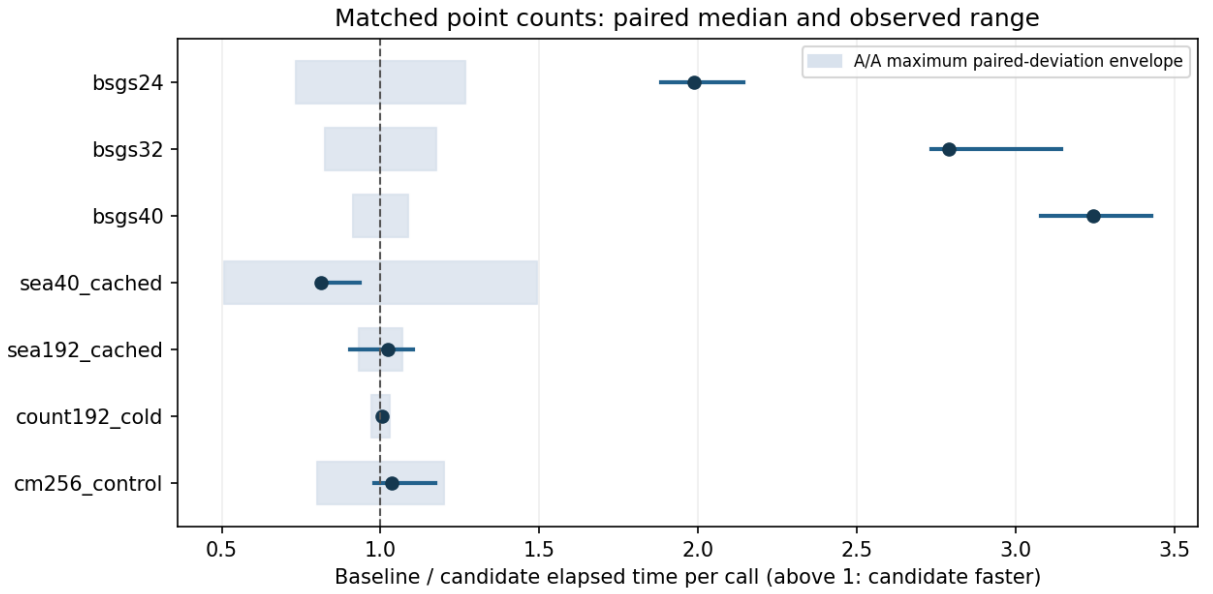
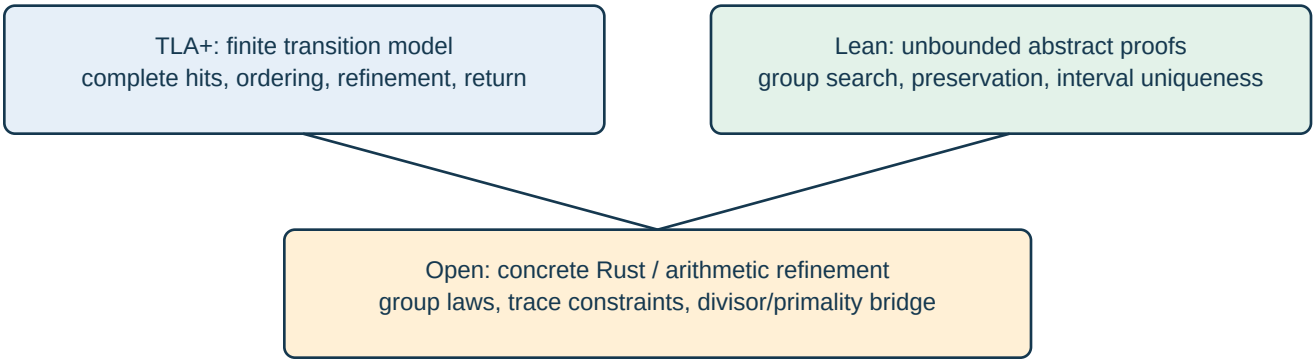
Point counting: measured changes and formal scope





2026-10-16; PR #1923. This report supplements the historical implementation study, which made no timing claim. The present study measures the original source cf274521 against the optimized source 44f8d17f, using identical native workers and verified orders. All 140 requested measurement records completed; none was excluded by the visible-process contention policy.

Observed performance

The three BSGS point-count workloads have paired median baseline/candidate ratios 1.98652, 2.79019 and 3.24310. Each exceeds its workload's observed A/A deviation envelope. SEA and the complete P-192 API do not establish an improvement under that predeclared noise policy. Cached SEA at 40 bits was slower in every A/B pair; its median ratio is 0.81228, within a particularly wide 49.308% A/A envelope. That observation is retained rather than hidden.

The following table is copied from the native summary. Times are measured microseconds per call, with five A/B pairs per workload. Ratios are medians of paired ratios, not ratios of the two displayed medians.

Workload	A median us/call	B median us/call	A/B median ratio	Paired range	A/A max deviation	A/B pairs
bsgs24	61.747	31.495	1.98652	1.88259..2.14328	26.575%	5 / 5
bsgs32	327.251	117.994	2.79019	2.73356..3.14247	17.594%	5 / 5
bsgs40	1936.899	591.519	3.24310	3.07922..3.42764	8.710%	5 / 5
sea40_cached	264.650	314.525	0.81228	0.80232..0.93396	49.308%	5 / 5

Workload	A median us/call	B median us/call	A/B median ratio	Paired range	A/A max deviation	A/B pairs
sea192_cached	1904284.547	1825125.454	1.02393	0.90254..1.10098	6.766%	5 / 5
count192_cold	98449066.339	97670607.289	1.00469	0.99858..1.01812	2.869%	5 / 5
cm256_control	19048.878	17948.105	1.03405	0.98103..1.17123	19.957%	5 / 5

The host is an AMD EPYC 9V74 VM with five visible CPUs. Children were pinned to CPU 4, movable threads were evicted, and visible-process CPU use was monitored under the repository benchmark lock. PSI is unavailable; some idle system threads could not be moved, and cloud neighbors and frequency were uncontrolled. This is not an exclusive-host measurement. The A/A envelope is a descriptive repeatability control, not a confidence interval.

The cached rows exclude canonical Phi-cache import and one warm-up call; their setup costs are retained separately. The original cache preparation recorded 82.296 seconds of setup. The cold P-192 row includes fresh polynomial construction, counting and curve/twist certification. Warm-stage timing is not substituted for complete API timing. The secp256k1 row is an unchanged CM point-count control; it does not time CM curve generation.

Algorithm and correctness changes

The implementation retains every baby index for a repeated group point, including repeated infinity, uses exact integer sizing and bounds the final giant block. Later sampled points can test only surviving candidates when the scalar-test cost estimate is lower than a complete search. Candidate intersections use ordered results. The small-field count also hoists the fixed giant-step multiple. These are engineering changes to the existing algorithms; no asymptotic complexity improvement is asserted.

The TLA+ sample-only negative control prompted a concrete constructor regression. For CM data p=11, D=-7, v=2, t=-4, the old constructor returned an order-8 curve when the requested order was 16. For p=59, D=-23, v=2, t=-12, it returned order 48 instead of 72. Both failed for all eight tested seeds. It also accepted invalid norm data t=-36 at p=59. The retained pre-fix run has 17 failures among 32 signed-trace/seed cases plus one invalid-input check.

The constructor now checks probable primality and the ordinary CM norm t^2 - D*v^2 = 4p before constructing a curve. For fields below 2^22 it compares an exhaustive count; for larger fields it requires successful curve and twist Hasse certificates. If neither twist is verified, it returns an error, including when factorization is insufficient. It exposes the verification evidence on CmCurve. The post-fix run passes all 33 checks. A 40-bit integration test exercises the certificate path for both trace signs, as well as the existing CM isogeny-loop check.

CM generation was already implemented for ordinary, nonexceptional j invariants through Hilbert class polynomials and twist selection. This patch repairs its order guarantee; it does not add general support for j=0 or 1728, prove the analytic polynomial rounding, or certify the full endomorphism ring. The constructor's GLS metadata h and r are not certified by its base-field order check. These scope limits are also stated in the API usage guide. The constructor fix happened after timing and is not included in either frozen benchmark arm. No generation speedup is claimed.

Checked mathematical argument

Let the lawful additive group have step P and target Q. For positive baby width B, every index k has the unique decomposition k=iB+j with 0<=j<B, by quotient and remainder. The scalar equation [k]P=Q is equivalent to [j]P=Q-[iB]P by addition and cancellation. A complete baby multimap therefore finds every solution in 0<=k<C, including repeated group points. Testing k<C excludes excess entries in the last block. Conversely every emitted match satisfies the scalar equation. This is Lean's bsgs_complete theorem, with smul_add and subtract_left proved from the stated group laws.

Within one block, insertion order increases j. Every valid index of an earlier block is smaller than every index of a later block, since iB+j < (i+1)B <= i'B+j'. A positive signed trace stride preserves this ordering. These are giant_blocks_ordered and signed_progression_ordered; the concrete table's insertion and traversal remain implementation obligations.

If the true candidate starts in C and every sampled test T accepts it, intersection C with T retains it. Induction proves this for any number of samples. Directly filtering C by T and intersecting a complete search for T are the same predicate. A singleton surviving set must consequently contain the true candidate. These are refinement_preserves, refinements_preserve, direct_equals_full_intersection and unique_sound. The premise that real SEA constraints and point tests accept the true trace is explicit, not proved here.

Finally, if two numbers lie in the same interval of width w and both are multiples of a genuine divisor d>w, they are equal: unequal multiples differ by at least d, contradicting the width. This is interval_unique, the uniqueness step of the Hasse certificate. Establishing genuine group-order divisibility and the arithmetic/primality bridge remains a separate obligation.

Formal execution and limits

TLC checked Fixed (165,750 distinct states), Wide (310,500), and Certification (58,956), including invariants, ordering and bounded termination. The legacy single-index table and sample-only certificate controls both failed with the expected nontrivial counterexamples and exit status 12. Lean 4.19.0 checked all ten theorems without admitted proofs. Axiom inspection reports only standard propext and Quot.sound where used. All source, pinned tool hashes, finite bounds and final logs are retained.

These are finite-state model checks and unbounded abstract lemmas, not a machine-checked refinement of compiled Rust. The proof map names the exact assumptions and open implementation obligation: prove the actual field/point operations form the group used by Lean, then prove SEA residue generation and the divisor/primality connection. Analytic CM construction, integer bounds, compiler behavior and timing resistance are also outside the checked claims. An independent lean4checker executable is unavailable; ordinary Lean kernel checking and axiom inspection completed.

Requirement status and reproducibility

Requested work	Status	Evidence or exact remaining gap
Improve point counting and SEA implementation	Implemented; executable checks pass	Exhaustive helper tests, actual curve regressions, release API/SEA/kernel tests; speed claims limited to measured BSGS cases
Determine CM generation support and implement if absent	Existing support audited and corrected	Signed-trace regression; exhaustive or Hasse order evidence; exceptional j and analytic certification remain unsupported/unproved
Measure speedup	Complete for all seven frozen workloads	140 records, five A/A and five A/B pairs each, matched orders/work, no excluded pairs
Model in TLA+ and check proofs	Checked within documented scope	Three positive TLC configurations, two negative controls, ten Lean lemmas, pinned CI replay
Full cryptographic implementation verification	Partial; open	Concrete Rust/group refinement, SEA congruences, certificate divisibility/primality and CM analytic proof are missing

Release validation passes 41 tests, with one pre-existing ignored test. The native analyzer also tests rejection of bad/mismatched records. Native and library Clippy checks, touched-file formatting, and the two CM example builds are recorded in evidence. The existing crate-wide formatting failure in lib.rs predates this branch; it is not hidden by editing unrelated code.

Replay commands are in README, measurement accounting and the run report. The chart source is render_reports.py, plotting only the native summary; both SVG and 150 dpi PNG copies accompany the PDFs under benchmarks/20261016-sea-point-count. The historical SEA report, performance-gains data, IC scoreboard/timeline and curve-transfer figures are unchanged: this point-count result does not alter their DLP workloads or recorded claims. No DLP or curve-security claim is made.